

Wavelet CT

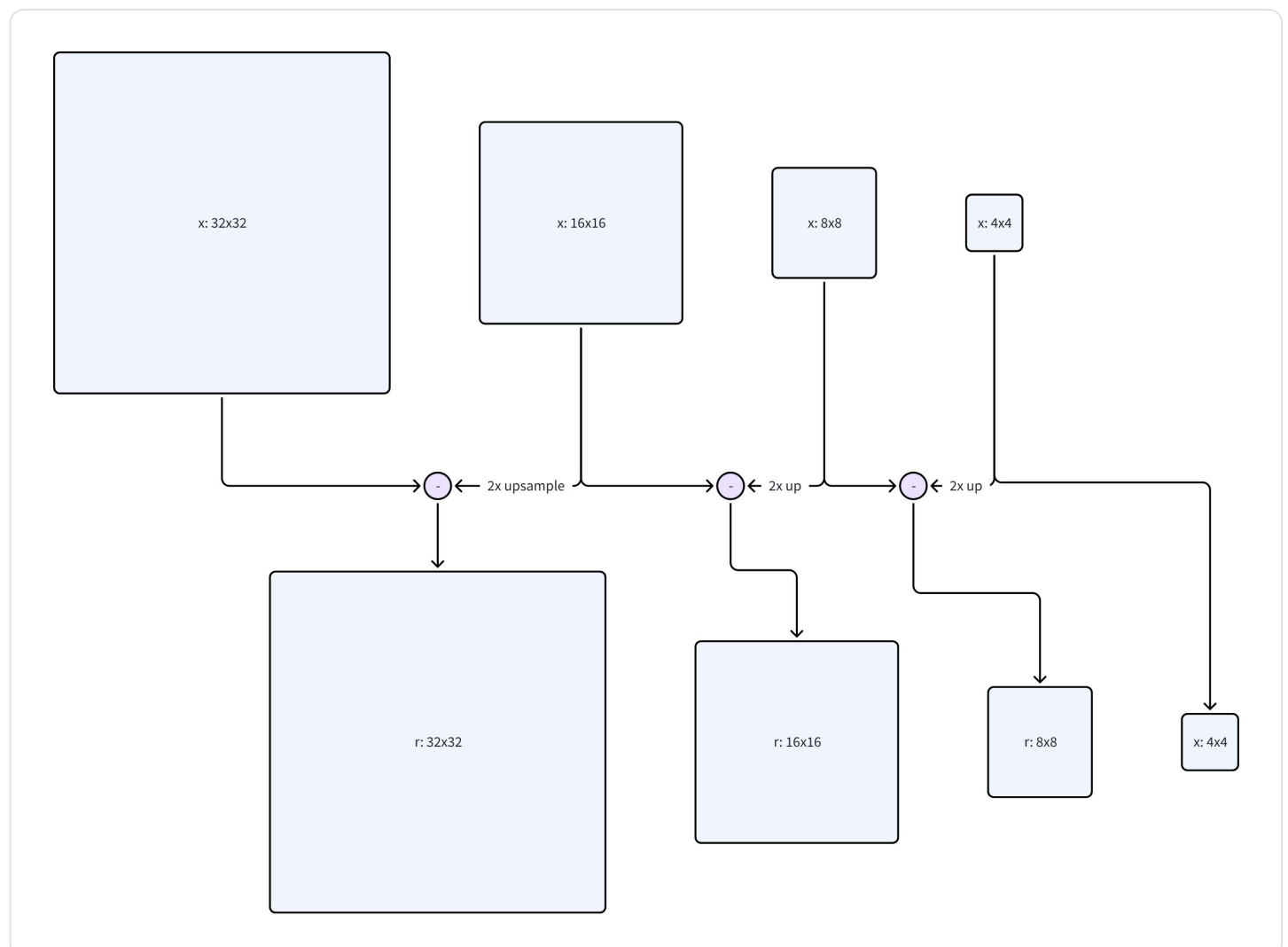
Observation:

- In the sampling process, diffusion models tend to generate low-frequency information first, and high-frequency information later
- In other words, high-frequency information depends on low-frequency information. 小的物件/细节纹理，长在大的物件上面。

Question:

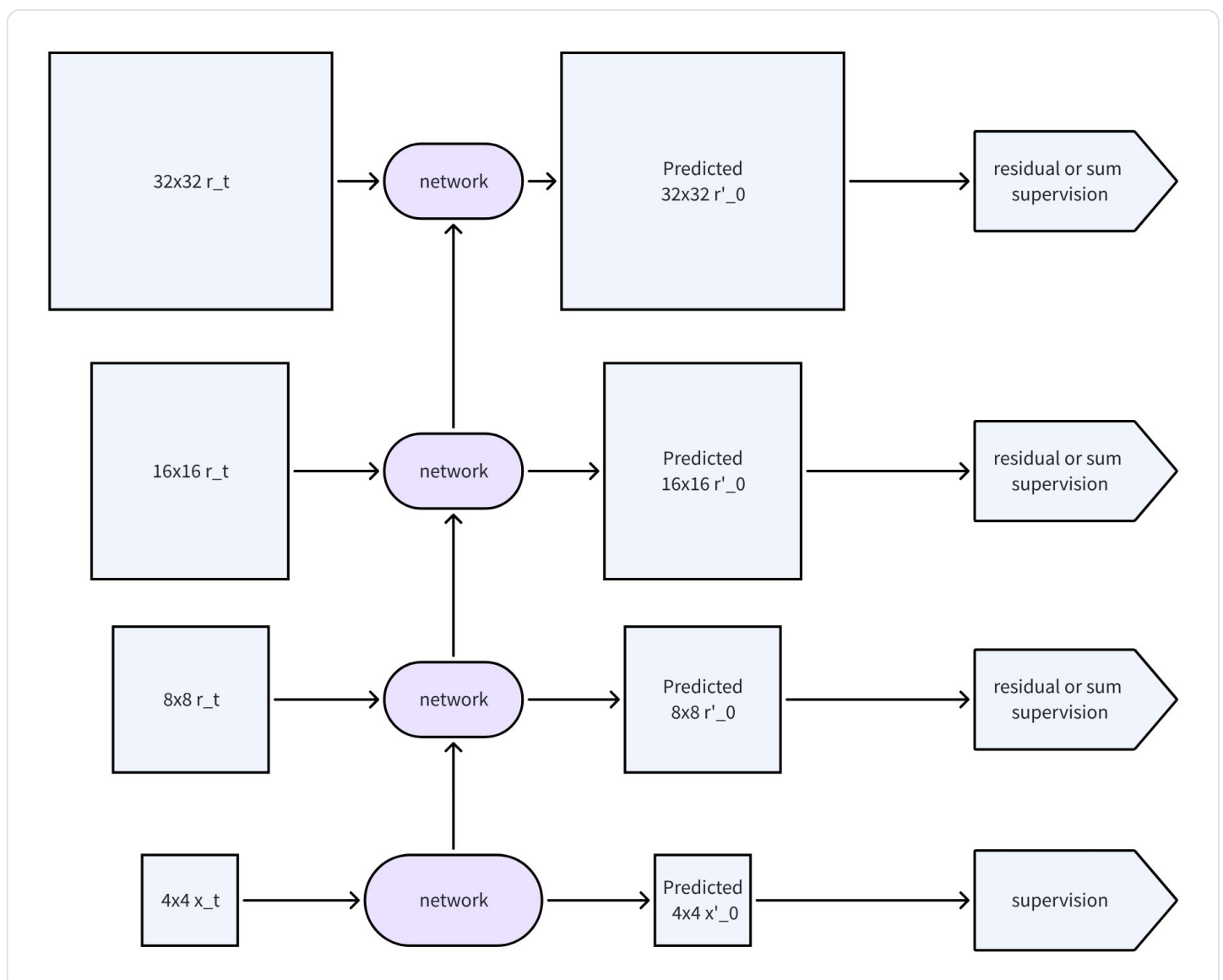
- Can we use this observation to enhance the performance of few-step methods such as CT/MF/GAN?

Method:



- Decompose an image x into a sequence of wavelet-transformed layers
 - Layer 0: 4x4, global average image
 - Layer 1: 8x8, residual image
 - Layer 2: 16x16, residual image
 - Layer 3: 32x32, residual image
- In the generator network, condition the generation of high-frequency layers on the outcome of low-frequency layers
 - Layer 0: 4x4, not conditioned on anything else
 - Layer 1: 8x4, conditioned on layer 0
 - Layer 2: 16x16, conditioned on layer0 + layer1
 - Layer 3: 32x32, conditioned on layer0 + layer1 + layer2

Generator structure:



- Final generated x_0' output is computed as $(4 \times 4 \ x'_0) + (8 \times 8 \ r'_0) + (16 \times 16 \ r'_0) + (32 \times 32 \ r'_0)$

Supervision approaches:

- Sum supervision: apply supervision on the final, summed x_0'
- Residual supervision (teacher forcing)
 - Apply supervision separately on $4 \times 4 \ x'_0$, $8 \times 8 \ r'_0$, $16 \times 16 \ r'_0$, $32 \times 32 \ r'_0$
 - Using target image's wavelet-transformed layers as groundtruth